

Group Log

Aura Coach: AI-Powered Real-Time Communication Trainer

DES646 Course Project

Indian Institute of Technology Kanpur

1. Introduction

Aura Coach is an AI-driven web application designed to provide real-time feedback on users' body language, vocal tone, and speaking performance. The project's objective is to help users enhance their communication confidence by combining computer vision, speech analysis, and emotion recognition — all in one interactive platform. Through an adaptive live dashboard, the system measures posture, facial emotion, tone, and speaking pace, offering instant feedback during a training session.

2. Initial Research and Ideation

The idea originated from the observation that many individuals struggle with posture, tone, and delivery during public speaking or interviews. Our goal was to build an accessible AI platform that acts like a “smart mirror” — analysing live posture, expressions, and voice in real-time.

The team explored multiple approaches:

- **Pose Estimation:** for detecting posture correctness and head tilt.
- **Facial Expression Analysis:** to estimate emotional state.
- **Speech Metrics:** to track speaking pace (WPM) and filler words.
- **Audio Signal Processing:** for tone and volume estimation.

We surveyed existing solutions such as *Yoodli* and *Orai*, finding that while they offer feedback post-recording, none provided live, multimodal analysis. This inspired us to build a tool that merges **visual, vocal, and emotional analytics** in a continuous feedback loop.

3. Design and Planning

Aura Coach was divided into three main components:

3.1 Frontend (React.js)

The frontend includes three main views:

1. **Home Page:** The entry interface featuring the project tagline “*Face. Train. Succeed.*” with a glassmorphic background.
2. **Training Page:** The core module where webcam and microphone access enable real-time analysis of posture, emotion, and voice.
3. **Report Page:** Displays the session summary, graphs, and AI-generated recommendations.

The UI design follows a minimalist aesthetic with gradient backgrounds, frosted-glass cards, and smooth transitions. Each panel — Body Language, Voice Style, and Feedback — updates dynamically during the session.

3.2 Backend (FastAPI – Python)

The backend was developed using FastAPI to:

- Handle data exchange between the frontend and AI modules.
- Stream live audio data via WebSockets for real-time analysis.
- Generate JSON-based reports with metrics and insights.

Audio signals were processed using **Librosa** to compute pitch, RMS energy, and tone categories such as *Calm*, *Balanced*, or *Energetic*.

3.3 AI and Model Integration

- **MediaPipe Pose** for shoulder and head alignment detection (posture scoring).
- **Face-API.js** for emotion recognition directly in the browser.
- **Web Speech API** for speech recognition, counting filler words and calculating WPM.
- **Librosa** and **NumPy** for backend audio feature extraction.

The combination of these tools allowed Aura Coach to deliver a cohesive multimodal experience — visual, auditory, and linguistic cues fused into one system.

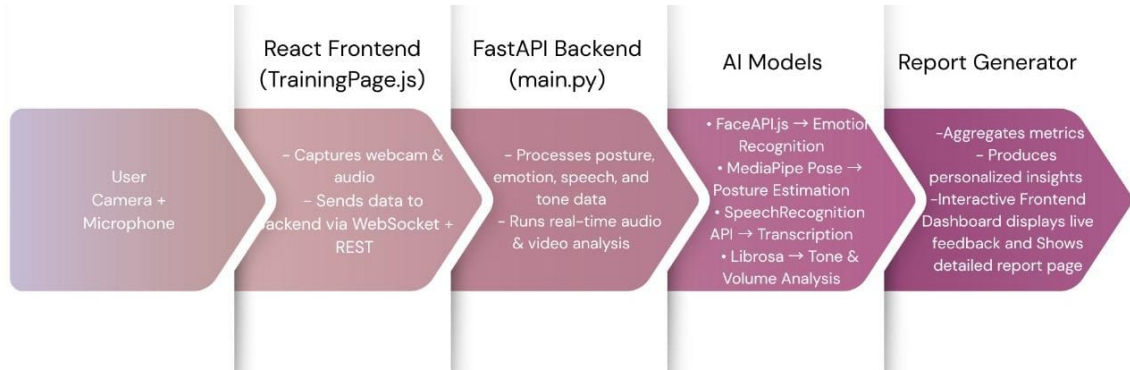


Figure 1: System Architecture of the Aura Coach platform showing the interaction between the React frontend, FastAPI backend, AI models, and report generation module. The architecture enables real-time emotion, posture, and speech-based feedback.

4. Working Notes and Implementation

4.1 Phase 1: UI and Frontend Setup

We built the interface using React components and modular CSS. The homepage animation used dynamic text transitions to display the tagline. The training dashboard was the most complex, integrating:

- Live webcam feed with pose detection overlays.
- Facial emotion tracking every second.
- Real-time speech-to-text analysis for word count and filler detection.
- Audio streaming to backend for tone and volume estimation.

4.2 Phase 2: Backend Communication and Logging

Each session logs metrics (posture, tone, emotion, WPM, fillers) to a unique JSON file. FastAPI's REST and WebSocket endpoints handle:

- `/api/session/log` for storing final session data.
- `/api/report/{session_id}` for generating summary reports.
- `/api/audio-stream` for real-time voice analysis.

4.3 Phase 3: Report Generation

At the end of a session, the user is directed to the Report Page, which fetches insights from the backend. A line chart (built using Chart.js) visualizes “Confidence Trend” over time, while textual insights summarize:

- Speaking speed and tone.
- Posture and emotional balance.
- Personalized improvement tips.



Figure 2: Workflow of the Aura Coach system, illustrating the end-to-end process from session initiation to performance report generation through real-time video, audio, and AI-based feedback.

5. Testing, Debugging, and Iterations

The project underwent several iterations to improve accuracy and usability:

- Optimized emotion model load times and posture detection frame rate.
- Fixed audio latency and improved synchronization between metrics.
- Refined the tone estimation logic to prevent false high-energy readings from background noise.

- Adjusted filler detection for short pauses in speech.

Cross-browser testing was conducted on Chrome, Edge, and Firefox to ensure microphone and webcam compatibility.

6. Results

The final version of Aura Coach successfully achieved:

- Real-time multimodal analysis (video + audio + text).
- Smooth live feedback through the interactive dashboard.
- Automated session summary with insights and visual graphs.
- Stable backend operation with concurrent session handling.

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The user interface and live demonstration of the web application are showcased during the final project presentation, highlighting the interactive dashboard and session analysis features.

7. Conclusion and Future Scope

Aura Coach demonstrated how AI and design can converge to create meaningful real-time learning tools. By integrating multiple sensory analyses into one system, the project helps users build self-awareness and improve communication quality.

Future extensions may include:

- Multi-user progress tracking and cloud-based session storage.
- Enhanced emotion detection using voice sentiment models.
- Integration with wearable devices for heart rate and stress tracking.
- Personalized AI assistant for real-time coaching feedback.